GEK-83789D

DRIVE SYSTEM INSTRUCTIONS

AC TO DC POWER CONVERSION AND CONTROL

for

DRESSER
CONSTRUCTION & MINING EQUIPMENT
MARION DIVISION

ROTARY BLAST HOLE DRILLS

DRIVE SYSTEMS OPERATIONS
SALEM, VIRGINIA 24163

GENERAL ELECTRIC

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SALEM, VIRGINIA, U.S.A. 24153

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SAFETY SYMBOL LEGEND

WARNING	Commands attention to an operating procedure, practice; condition, or statement, which, if not strictly observed, could result in injury or death.
CAUTION 1	Commands attention to an operating procedure; practice; condition, or statement, which, if not strictly observed, could result in damage to, or destruction of equipment.
NOTE	Commands attention to an essential operating or maintenance procedure, condition, or statement which must be highlighted.



THIS EQUIPMENT CONTAINS A POTENTIAL HAZARD OF ELECTRICAL SHOCK OR BURN. ONLY PERSONNEL WHO ARE ADEQUATELY TRAINED AND THOROUGHLY FAMILIAR WITH THE EQUIPMENT AND THESE INSTRUCTIONS SHOULD INSTALL, OPERATE, OR MAINTAIN THIS EQUIPMENT.

ISOLATION OF TEST EQUIPMENT FROM THE EQUIPMENT UNDER TEST PRESENTS POTENTIAL ELECTRICAL HAZARDS. IF THE TEST EQUIPMENT CANNOT BE GROUNDED TO THE EQUIPMENT UNDER TEST, THE TEST EQUIPMENT'S CASE MUST BE SHIELDED TO PREVENT CONTACT BY PERSONNEL.

TO MINIMIZE HAZARD OF ELECTRICAL SHOCK OR BURN, APPROVED GROUNDING PRACTICES AND PROCEDURES MUST BE ADHERED TO.

****
* CAUTION *
****

THIS EQUIPMENT IS DESIGNED TO WORK IN A NORMAL INDUSTRIAL ATMOSPHERE. OPERATING RADIO TRANSMITTERS NEARBY MAY CAUSE THE CONTROL EQUIPMENT TO MALFUNCTION. IT IS RECOMMENDED THAT RADIO TRANSMITTERS NOT BE OPERATED WITHIN THE VICINITY OF THE CONTROL EQUIPMENT WHEN THE PANEL DOORS ARE OPEN, NOR AT ANY TIME WITHIN THE CONFINES OF A POWER CONTROL ROOM.

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-- NOTES --

-- SECTION 1 --
INTRODUCTION

l-l. Purpose

This instruction book provides information concerning the installation, tune-up, operation, and maintenance of General Electric Company's AC to DC static power conversion and control equipment designed for the electric rotary motion drive of Marion Division blast hole drills.

These instructions do not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met during installation, operation, and maintenance. Should further information be desired or should particular problems arise that are not covered sufficiently for the purchaser's purposes, the matter should be referred to General Electric Company, Drive Systems Operations, Salem, Virginia, U.S.A.

1-2. Audience Level

This instruction book is written at a technical level suitable for experienced electricians and technicians.

-- NOTES --

-- SECTION 2 --
DESCRIPTION OF EQUIPMENT

2-1. FUNCTIONAL DESCRIPTION

The General Electric control panel for Marion Division rotary blast hole drills provide the following functions:

- AC to DC static power conversion
- Adjustable regulated armature voltage
- Tapered current limit
- Adjustable regulated motor field excitation
- Protection against instantaneous overcurrent, phase loss, armature ground, motor field loss, and motor field overcurrent
- Transition of control from the remote driller's rheostat to preset low speed or collaring speed setting
- Transition between torque (motor field current) level as selected by the remote torque selector switch

2-2. PHYSICAL DESCRIPTION

Enclosure: Extremely rugged NEMA 1
Dimensions: Approximately 76 in. high, 44 in. wide, 28 in. deep
Weight: Approximately 1800 lbs.

For further details, refer to outline drawing 206B6723 and assembly drawing 228B7236.

2-3. ELECTRICAL DESCRIPTION

Incoming Power

These rotary blast hole drill drive systems are supplied from either 380 or 435 volt 50 Hz or from 460 or 600 volt, 60 Hz systems. On 380 and 600 volt systems, buck-boost transformers are used to transform incoming power to approximately 460 volts. Refer to the Marion Power Shovel company drawings and specifications for actual connections and ratings.

DC Voltage Settings

Maximum no-load DC voltage output should be set for 525 volts with 435 volts input to the converter. The maximum no-load DC voltage setting

corresponding to 460 volts input should be 550 volts.

Converter DC Output Current
For actual current ratings of the drive motors and converters, refer to the Marion Division drawings and specifications.

Drive Settings
Settings should be made in accordance with the data sheets included in elementary 228B7235 sheets 1J (Rev. 4), 1K and 1L (Rev. 2).

2-4. REQUIRED TEST EQUIPMENT

These instructions require the following test equipment:

2 - Digital multimeters

1 - 60A-50mV shunt for checking motor field current.

6 - Jumpers

2 - Sets of meter leads

1 - Digital meter capable of reading low ohms

2 - 200A - 50mV shunts for measuring armature current (2-motor drive only.)

-- SECTION 3 --
START-UP PROCEDURES

3-l. INITIAL INSPECTION

This section lists the steps that should be taken for initial inspection of the equipment.

NOTE: Power is turned off when executing the procedures in this section.

3-1.1. Open the armature loops by disconnecting the cable from the LE contactor to the DCP terminal's power conversion bridge.

3-1.2. (1) Check the AC power circuits for shorts and grounds. (See elementary diagram 228B7235 sheet 1C of AC power distribution.)
(2) Check the DC power circuits for shorts and grounds.
(3) Check the ground protection circuit: read resistance to ground, and check the circuit to make sure it is there.

3-1.3. Load sharing resistors are supplied on two-motor drive systems only; The resistance connected in series with each armature circuit should be approximately 0.11 to 0.13 ohms cold. With the two motors stalled and their fields disconnected, apply about 8 volts to the parallel armature circuits; Using shunts in series with each motor, measure and record the two currents. Do not allow current to flow for more than 5 or 6 seconds. If the currents are not approximately equal, increase the resistance in series with the motor with the highest current and take a new measurement of armature currents. Adjust the resistance to achieve a close balance between the two motors (within 5% of each other).

3-1.4. (1) Check the motor field resistance by measuring resistance from Field Positive to Field Negative on motor field exciter's TB2-1 to TB2-3. Resistance should be approximately 8 ohms (cold) for 2 motors and 16 ohms (cold) for 1 motor,
(2) If the machine's main transformer exceeds 350 KVA, check that the reactors are in series with the main breaker's incoming lines. Note: Buck-boost transformer may offset need for reactors.
(3) Check the jumpers on the field exciter daughter board.

3-1.5. Check the reference rheostat in the operator's cab. The wiper should be on the ACOM end of the pot with the pot set for "0" speed reference.

3-1.6. Check the torque connection switch for the proper connection per the elementary diagram sheet 1A.

3-1.7. Check the stop/start circuit for shorts and grounds.

3-1.8. Open the SCR bridge's front cover and inspect for shipping damage and loose material.

3-1.9. (1) Check all control plugs and ribbon cables to ensure that they are tight (be sure to check the plugs behind the DS3800PS1's hinged door.)
(2) Check wiring for the remaining AC power circuits per elementary diagram sheet 1C.

3-1.10. On the DS3800NFOA card, check the following:

For 60 Hz power, connect SEM pin 72 in slot B (NFOA) to ACOM. For 50 Hz power, no jumper is required. See elem. sheet 2F.

3-1.11. Adjust the pots (SEM slot C) on the DS3800DSVA daughter board (DSVA), located on the DS3800NSVA board (NSVA), as follows:

R1 = 25K/50K (midpoint)
R3 = 3.6K/10K from the pot wiper to the pot terminal not connected to the 2.74K resistor on the DSVA.

Check the following fixed resistor/capacitor values on board:

C10 - None	R15 - None	R19 - 10K
C18 A - None	R16 - 200K	R23 - 3.32K
C18 B - None	R18 - None	R24 - None

3-1.12. Check resistor/capacitor values on the DS3800NHVA/DHVA High Voltage board, mounted on the front of the SCR bridge. (Refer to elementary diagram sheet 1K.)

3-1.13. Check the following resistor/capacitor values on the DS3800DRMA daughter board (DRMA), located in module Slot C:

R108 = 150K
R109 = 100K

3-1.14. Insert a 60A-50mV shunt in the motor field circuit. The shunt may be mounted on the MPS terminal board for motor fields in the top of the power conversion module (PCM) compartment.

3-1.15. Pull the fuses on the motor field exciter (FE), then insert a temporary jumper between the FE-TB8 (8) wire 1A47G and the TB8 (12) wire 1A47K to permit the LEX relay to pick up with the field off. Check Berg jumpers on the FE daughter board for the correct setting:

- For 50 Hz operation, connect daughter board pin DA11 to ACOM
- For 2 motors, insert Berg jumpers YA-YC and YB-YD
- For 1 motor, insert YA-YC only

CAUTION

DO NOT PLUG IN OR UNPLUG ANY BOARD WHEN POWER IS ON.

3-2. MAGNETIC DEVICE OPERATION

This section lists the steps for operating the magnetic device.

NOTE: Power is turned on when executing the procedures in this section.

3-2.1 Close both the PCM feeder breaker (located in the AC cabinet) and the control circuit breaker (located in the PCM) to energize the PCM.

3-2.2. Check the lights on the NHVA board: all cell lights (neon) should be on; all firing pulse lights (red) should be off.

3-2.2. Confirm that the PCM fans (if applicable) come on.

3-2.4. Confirm that the lights on the PCM, the SEM, the relay module (RM), the fast dropout relay module (FDR), the power supply (PS), and the motor field exciter (FE) come on as follows:

LIGHT	LOCATION	INDICATION	COMMENT
6 neons	PCM-DS3800NHVA	Bridge energized	
6 LEDs	PCM-DS3800NHVA	Gate pulses	Disregard
Status	PS	Power supply volt correct	
Fault FDR		Fast dropout relay reset	
SFLTX	RM-IC3600LING	No summation fault	
Status	PS	Power supply volt correct	
ZVX	RM-IC3600LING	PCM near 0 volts output	
PLL	SEM-DS3800NFCB	Phase-locked loop locked in	
UNSUP	SEM-DS3800NFCB	Firing circuits unsuppressed	
IMOK	DS3800NFOA	NFOA board operational	
SEQ	DS3800NFOA	Correct phase sequence	
PULS	DS3800NFOA	Correct 6 SCR pulses	
TOF	DS3800NFOA	Disregard	
TOR	DS3800NFOA	Disregard	
ACF	DS3800NFOA	AC DC	
DCF	DS3800NFOA	DC AC	
PCA	DS3800NRMA	Disregard	
IMOK	DS3800NSVA	Regulator amplifiers unsaturated	

3-2.5. Place the torque selector switch in the neutral (central) position and operate the start switch, both at the main console. Confirm that the LEX and LETDX lights on the RM come on, that loop contactor LE picks up, and that the motor blower (if applicable) comes on.

3-2.6. Make sure that the speed pot (reference) is in the zero speed position. Then move the torque selector switch to the forward low and high torque positions, and the reverse low and high torque positions, and confirm that the FX, FSX, and RX relays operate as indicated by lights on the RM.

3-2.7. Confirm that each stop pushbutton correctly de-energizes immediately the LEX relay, and after approximately a 1 second delay, the LETDX relay and the LE contactor.

3-2.8. Check the DS3820PS1 power supply voltages on the SEM as follows:

- o TP4, P15 - ACOM = +15V ±.05 Adjusted by pot R2
- o TP5, N15 - ACOM = -15V ±.05 Adjusted by pot R3
- o TP1, P28 - DCOM = +28.0V ±.70 Adjusted by pot R5
- o TP2, P5 - DCOM = +5.0V ±.05 Adjusted by pot R2
- o TP6 = ACOM
- o TP3 = DCOM

3-3. OVERCURRENT, SUPPRESSION, AND FAST TRIP CHECK

This section lists the steps for checking overcurrent, suppression, and fast trip.

NOTE: Power is turned on, open loop when executing the procedures in this section.

3-3.1. Remove and temporarily replace R41 and R42 Burden resistors on the DHVA board with higher value resistor, such as 7 to 10K ohms, to prevent the DS3820TRMA Regulator Test Module (RTMA) suppression test circuit from loading down.

3-3.2. Connect (jumper) NSVA pin 42 to ACOM.

3-3.3. Connect the RTMA provided with the panel into the SEM's jack JL. (Alternatively, the DS3820DTMA Drive Test Module may be used; however, the following instructions are written for the RTMA.)

3-3.4. Leave the motor armature circuit open as disconnected in section 3-1.1.

3-3.5. Connect a digital voltmeter or equivalent to measure the voltage at test jack REF2. (Measure DC control voltages with respect to ACOM on the RTM.)

3-3.6. Turn on the power and use the RTM to turn on the SCR bridge in the forward or reverse directions by connecting the RTM's RF1 output to TP8 on the NSVA board. Read the volts on the *NSVh's* TP9 (+ volts = forward direction). Turn the RTM's TEST switch to the ON position with slight t voltage on the *NSVh's* TP9. Apply enough volts on the *NSVh's* TP8 to cause the NHVA board's red LEDs to come on in the forward direction (upper set of LEDs). Raise the bridge volts to 150 to 200VDC.

3-3.7. With the SCR bridge turned on to approximately 150 to 200VDC in the forward direction, apply approximately +2VDC to the RTM's reference test jack.

Push the suppress switch down to dump the +2V through the 7 to 10K Burden resistor.

No trip should occur (red LEDs stay on). If a trip does occur (red LEDs turn off or pulsate), turn off the RTM's TEST switch. Then turn the DS3800DFCB's forward suppress pot R3 clockwise to raise the trip level; (The DFCB is located on the DS800NFCB board, which is in module slot A).

Repeat the above procedure for increasing steps of trip voltages on the RTM's test jack REF2. (Example: 2, 4, 6, 8, 10.)

Suppression level is set correctly when approximately +9.5V *on* REF2 causes the red LEDs to go out or pulsate (logic overcurrent occurs at 80% of trip level).

Increasing the REF2 voltage to 11.75V should cause suppress operation--i.e. the NFCB board's UNSUP light should turn off. The FDR module's fault light should turn off, and the summation fault relay SFLTX, the line excitation relay LEX, and the loop contactor LE should instantaneously drop out to suppress all firing pulses (red LEDs), to suicide the regulator, and to open the armature loop (if the loop was not already open for test purposes).

With 11.5V on REF2, the voltage on the RTM's suppress jack should momentarily jump to 11.25V when the suppress switch is closed and suppression occurs.

NOTE: 11.75V = 11.25V across the Burden resistor +.5V diode drop between test jack REF2 and RTM capacitors that act as a suppression current source.

NOTE: The LE contactor will only drop if it has been picked up by picking up excitation.

3-3.8. Check suppression in the reverse direction by repeating the above steps. However, these steps will differ in section 3-3.6, in which the volts on the NSVA board's TP9 should now be negative, and the reverse LEDs (lower set) on the NHVA board should come on. The reverse suppression pot is R7 on the DS3800DFOA board (located on the DS3800NFOA in module slot B).

NOTE: Turn the suppression pot counterclockwise (CCW) to lower the trip point, and clockwise (CW) to raise the trip point.

3-3.9. Remove power and replace the temporary Burden resistor with the correct resistor. Remove the jumper that connects the NSVA board's pin 42 to ACOM.

3-4. NO LOAD VOLTS ADJUSTMENT

This section lists the steps for making a no load volts adjustment.

NOTE: Power is turned on when executing the procedures in this section.

3-4.1. (1) Have an operator move the torque selector switch to the fast forward (torque selector #2) position and slowly increase the speed pot from zero volts output to 100% while monitoring the SCR bridge output volts.

(2) Adjust pot R2 on the DSVA board (located on the NSVA board in SEM slot C) for +525/550VDC on the SCR bridge output. Check reference at the NSVA's pin 49 for 10V. The NFCB board's VFB test point (TP13) should be -8.0V.

(3) The NFCB board's CFB test point (TP1) should be O.OVDC.

If the VFB test point exceeds 8.0VDC for 525/550VDC, adjust the reference on the NSVA's pot R2 to lower bridge volts to 475/525VDC, and adjust the DFCB's VFB pot for 8.0VDC at the NFCB's TP13, or until the SCR bridge volts reach 500/550VDC. Continue to adjust pots R2 and VFB until 525/550VDC = 8.VDC.

NOTE: If you cannot get 525/550VDC out of the SCR bridge for either the forward or reverse direction, adjust the voltage compounding pots: Forward direction - AUX 1 (R7) on the NSVA board; reverse direction - RH2 on the NCBA board.

To get voltage compounding out of the way at no load volts (see Figure 1 for the effect of voltage compounding), check the no load volts in the reverse direction. Make no adjustment and record the values.

3-4.2. After setting no loads volts, have the operator move the selector switch to position #1 forward, and set the output volts to 100VDC with the relay module's RHLS pot. Check the voltage level in #1 reverse and make no adjustment.

3-5. SET MOTOR FIELD EXCITER

This section lists the steps for setting the motor field exciter. Refer to elementary diagram sheets 1D for instructions, which will in most cases be the same as those in this section.

The FOC setting, 120% of FMAX, must be reset if the DS3800NFEX board (NFEX) or the field exciter is replaced. (See note in Section 3-5.3).

3-5.1. (Using pots on the motor field exciter), set the F-max pot full CW for the maximum current range. With the torque selector switch in neutral, and relays FX and RX de-energized, use pot F-min to set the field loss. The MPS setting for field loss may be lower than the lowest setting of the F-min pot. In this case, raise the field loss to trip the field at the lowest F-min setting. Mark the position of the field loss pot and raise the F-min setting by the amount of the difference between the field loss setting and the minimum field setting. Raise the field loss pot setting to trip unit, then mark the field loss pot. The difference between the 2 marks on the field loss pot is the amount the field loss pot should be rotated to below the minimum field set point. (See elementary diagram sheet 1J.)

Example:

F-min = 6 amps
Field loss = 4.5 amps
Lowest in setting = 5 amps

$$\text{Field amps (2 motor)} = \frac{\text{Volts at TP1}}{.35} \pm 15\%$$

$$\text{Field amps (1 motor)} = \frac{\text{Volts at TP1}}{.66} \pm 15\%$$

3-5.2. (1) Trip with field loss at the 5 amps, and mark the field loss pot.

(2) Trip with field loss at the 5A + (6 to 4.5A) = 6.5A, then mark the field loss pot.

(3) Trip field loss at 6 amps and rotate field loss back by the distance between the 2 marks on the field loss pot.

3-5.3. (Using pots on the relay module), put the torque selector in forward #2 (relays FX and FSC picked up), then set the field for full speed forward/reverse using pot RHH.

NOTE: Prior to setting the field for full speed setting, use pot RHH to increase the field to 120% of full field strength and set the field overcurrent pot. This pot is located on the DS3800NFEX board's bottom edge, 2/3 of the board length from the front-edge.

3-5.4. Put the torque selector in position #1 forward. Set the low speed forward field strength using the relay module's RHLF pot. (Only the FX relay is picked-up.)

3-5.5 Put the torque selector in position #1 reverse. Set the reverse field strength using the relay module's RHLR pot.

3-6. VOLT-AMP CURVE CHECK

This section lists the steps for checking the volt-amp curve.

NOTE: The equipment is open loop during these steps.

3-6.1. Move the torque selector switch slightly off neutral in the forward direction. Check the bridge output for the correct polarity.

3-6.2. Use an XY plotter to plot the volt-amp curve. (If one is not available, a number of points on the curve can be taken and plotted manually.)

3-6.3. Using a jumper, connect the RTM (REF1) test jack to the NFCB board's input current test point TP3 to inject a simulated armature current signal. A test signal of 7.5 volts corresponds to a stall setting of 200% armature current.

Sketch the desired volt-amp curve according to MPS settings (see Figure 1 for an example). Use the following test points:

SIGNAL	TEST POINT	LOCATION	SCALING
VFB	TP10	NSVA	-8V = Rated voltage
CFB	TP1	NFCB	-lov = Stall current

3-6.4. To set the VA curve, use the following steps:

(1) Have an operator turn the drive full on, and leave the control in this position until test is complete. 525/550VDC = 8.0VDC at VFB, and CFB = 0.

(2) Inject + volts via RTM ref #1 into the NFCB board's TP3 to simulate armature current. Adjust forward current limit bias pot FWDCL (R4 on NSVA) while injecting the signal into TP3 to set the CFB test point to 10.0VDC. The bridge volts and VFB should read as follows:

NO. OF MOTORS	BRIDGE VOLTS	VFB	CFB
2*	85VDC	1.22	10.0V
1	45VDC	.66	10.0V

* Bridge volts at stall = 85VDC on 2-motor units due to load sharing resistor volt drop for 600A stall.

For 415VAC input: $VFB = \frac{VDC}{525} \times 8V$ For 460VAC input: $VFB = \frac{VDC}{550} \times 8V$

(3) Pick a stall value close to midway in the voltage compounding portion of VA of Figure 1. Read the corresponding volts. Example: For a stall of 600A = 10V at CFB and a peak power point of 40% of stall with 2 motors. Set injected + volts via RTM to give 6.0V at CFB. Adjust the NSVA board's forward volt compounding pot AUX1-R7 until bridge volts = 400VDC and VFB = 5.8VDC.

(4) Recheck stall values and adjust if necessary.

(5) Recheck the voltage taper line in case stall values were adjusted too much.

(6) Plot a simulated VA for several current values (see Figure 1) around the peak power point to display the notch caused by the drive characteristic current limit.

(7) Be sure that the simulated current is at zero prior to reducing the bridge volts from full on.

(8) Repeat the procedure for the reverse direction and adjust the NCBA's RH2 pot.

3-6.5. Remove RTM. Set low speed volts at 100V or per MPS specifications by putting the drive torque selector switch in the #1 position forward and adjusting pot RHFS on the relay module. If applicable, operate the collaring interlock and set collaring bolts with pot RHC on the relay module.

3-7. VOLTAGE REGULATOR RESPONSE CHECK

NOTE: voltage regulator response (approximately 5 radians/second) has been calculated with the NSVA's voltage gain pot R3 at its midpoint (5K/10K ohms). Current limit response (approximately 200 radians/second) has been calculated with the NSVA's current limit gain pot R6 at its midpoint (10K/20K ohms). While it is impractical to try to check the current limit response with a brush recorder, the voltage regulator response can be checked as detailed below. (If a brush recorder is not available, you can omit this step.)

Apply a small step in the NSVA's TP2 and check the response at the NSVA's TP10 using a brush recorder.

3-8. CLOSED-LOOP STALL CURRENT CHECK

This section lists the steps for checking the closed-loop stall current.

3-8.1. De-energize the equipment and close the loop. Connect a meter to measure the current feedback signal CFB on the NFCB board's TP1. (10V = 200% rated current.) Chain off the drill stem.

3-8.2. Move the torque selector switch forward, and advance the driller's rheostat. Current should increase to 200% as indicated by -10V at the CFB test point.

CAUTION

DO NOT let stall current circulate for more than several seconds.

3-9. DRILLING OPERATION CHECK

The AC to DC power conversion and control unit is now ready for final check-out during the actual drilling operation. To do this, observe performance during all modes of operation to confirm that the unit is operating as intended.

-- SECTION 4 --
OPERATING INSTRUCTIONS

4-1. ENERGIZING THE EQUIPMENT

Close the power conversion panel (PCM) feeder breaker located in the AC cabinet to provide, 3-phase, 50/60 Hz power as applicable to the PCM. (In later equipment, this breaker resides in the PCM panel.) Close the control circuit breaker within the PCM panel to energize the control and motor field exciter circuits. The equipment should now be ready to start.

4-2. STARTING THE ROTARY DRIVE

To start the rotary drive, place the torque selector switch located on the main console in the neutral position. Operate the Excitation Start pushbutton on the main console.

4-3. SPEED AND TORQUE CONTROL

4-3.1. Collaring Operation

Closing the remote collaring interlock causes the drill to operate in the forward direction at the low speed preset by potentiometer RHC in the PCM.

4-3.2. Low Speed Forward/Reverse Operation

Place the torque selector switch in the forward-low speed or reverse-low speed position. The drill will run at the low speed preset by the PCM's pot RHLS. For low speed forward operating, the motor is excited at the field strength preset by the PCM's pot RHLF.

4-3.3. High Speed Operation

Place the torque selector switch in the forward-high speed or reverse-high speed position. Adjust the driller rheostat to control drill speed over full range. The motor is excited at the field strength preset by the PCM's RHH pot.

4-4. STOPPING THE ROTARY DRIVE

Stop the drill by moving the torque selector switch to neutral. Next, open the loop contactor using the excitation stop pushbutton on the main console.

4-5. DE-ENERGIZING THE EQUIPMENT

Open the PCM feeder breaker in the AC cabinet (or in the PCM) to remove, 3-phase, 50/60 Hz power from the equipment after the drilling operation is completed.

-- NOTES --

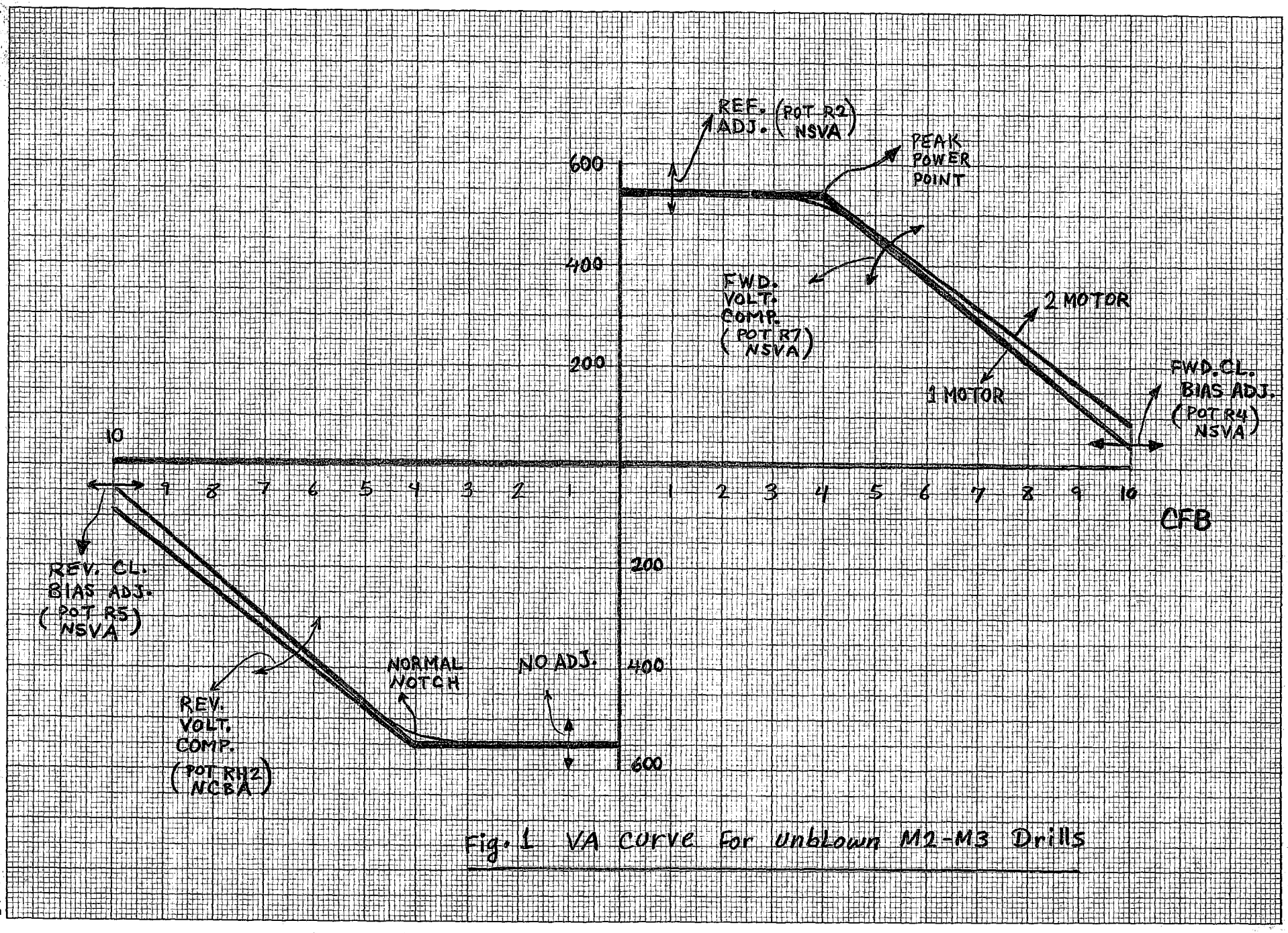


Fig. 1 VA Curve For Unblown M2-M3 Drills

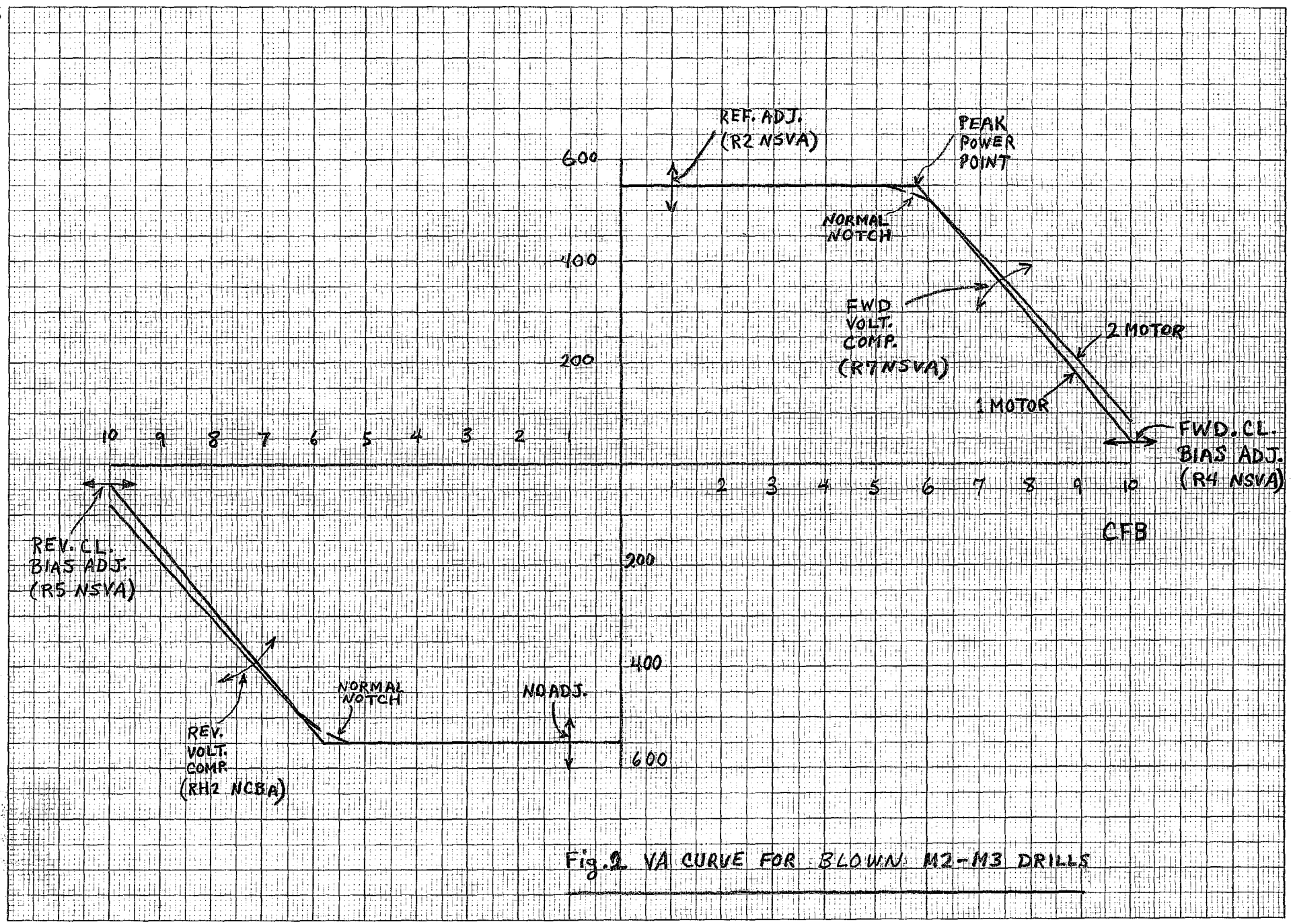


Fig. 2. VA CURVE FOR BLOWN M2-M3 DRILLS

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